



+

Business Requirements Specifications

For Servesense

Version 1.1

Project Overview	3
System Actors & User Roles	3
Role-wise User Journeys & Features	3
Epic 1. Waiter (Frontline User)	3
Story 1.1 Login & Access:	3
Story 1.2 Session Dashboard (Pre-Interaction):	4
Story 1.3 Live Session Experience	5
A. Live AI Suggestions & Recommendations:	
Data Captured by System (No Manual Input)	5
B. Tone & Behaviour Guidance (System Generated – Real-time)	5
C. Menu Recommendations Panel	6
D. Guest Feedback Capture	6
Story 1.4 Waiter Main Dashboard (Post-Session):	7
A. Performance metrics:	7
B. Conversation History:	8
C. Leaderboard View (Peer Comparison)	9
D. Learning & Coaching:	10
Waiter (Frontline User) Flow	11
2. Restaurant Manager (Admin / Manager Role)	12
2.1 Signup & Login	12
2.2 Staff Management	12
2.3 Restaurant Orientation & Data Setup (detailed down in 3. orientation epic)	13
2.4 Manager Dashboard (Business & Performance View)	13
A. ROI & Business Impact Summary (Read-only Metrics)	13



Date: 12/01/2026

B. Revenue & Sales Analytics	13
2.5 Staff Performance List & Drill-Down View	14
2.6 Knowledge & Coaching Management	17
Restaurant Manager (Admin / Manager Role) Flow	17
3.) Orientation (Data needs to be feed by the Manager)	18
Module 3.1: Restaurant Standard Policies (MANDATORY)	18
Module 3.2: Menu Knowledge (MANDATORY)	19
Module 3.3: Service SOP – Flow of Service (MANDATORY)	19
Module 3.4: Communication, Tone (MANDATORY)	20
Module 3.5: Best Practices & Excellence (ADVANCED)	21
Module 3.6 Sales Goals & Campaigns	21
4. System Output (AI & Platform Logic)	22
4.1: Conversation Processing Engine	22
4.2: Intent, Emotion & Tone Engine	22
4.3: RAG Knowledge Engine	23
4.4: SOP & KPI Evaluation Engine	24
4.5: Recommendation & Learning Engine	24
4. 6 Order Confirmation Prompts	25
A. Coaching Effectiveness (System Metrics)	26
B. Learning Progress & Adoption	26
C. AI-Recommended Actions	26
2.6 Overall Impact Summary	27



AI-Powered Real-Time Conversation Intelligence & Performance Management Platform for Restaurants

Purpose of Document

This Statement of Work (SOW) defines the scope, features, responsibilities, and high-level technical approach for designing and developing an AI-driven mobile and web-based system that assists restaurant waiters during live customer interactions and provides post-session performance analytics to restaurant management.

Project Overview

The objective of this project is to build an intelligent assistant platform that:

- Provides real-time, AI-powered suggestions to waiters during live customer conversations
- Ensures SOP compliance, accurate menu knowledge, and appropriate tone
- Analyzes conversation quality, sentiment, and performance
- Generates post-session summaries, KPI-based scoring, and coaching recommendations
- Enables restaurant managers to track performance, drive upsells, and improve service quality

The solution will consist of:

- A Flutter-based mobile application for waiters
- A web-based admin/manager dashboard
- An AI backend leveraging LLMs, speech processing, and Retrieval-Augmented Generation (RAG)

System Actors & User Roles

The platform operates with three primary roles, each having distinct journeys, responsibilities, and feature access:

- I. Waiter / Receptionist (Frontline User)
- II. Restaurant Manager (Operational & Performance Owner)
- III. Orientation (Data needs to be fed by the Manager)
- IV. System Output (AI & Analytics Layer)

Role-wise User Journeys & Features

Epic 1. Waiter (Frontline User)

Story 1.1 Login & Access:

User Stories: As a Waiter, I want to log in using my assigned credentials so that I can access my shift-specific dashboard & I should only see data related to my assigned outlet and shift.

Fields & Inputs

Field Name	Data Type	Mandatory	Validation	Source
Username / Employee ID	Text	Yes	Unique, case-insensitive	Waiter



Password / PIN	Password	Yes	Min length, masked	Waiter
Device ID	String	Yes	Auto-captured	System

System Behaviour

- Validates credentials using secure authentication.
- Applies role-based access control (RBAC).
- Logs login/logout timestamps.

Story 1.2 Session Dashboard (Pre-Interaction):

User Stories: As a Waiter, I want to start a session when I approach a table so the system can assist me live & I can pause a session if I step away temporarily & resume when I approach the same table and End session once the guest leaves.

The waiter accesses a simple session setup screen to initiate guest interaction. A session represents one table or guest interaction.

Fields & Inputs

Field Name	Data Type	Mandatory	Validation	Source
Guest Name	Text	No	Max length	Waiter
Table Number	Text / Number	Yes	Must be unique for active sessions	Waiter
Service Mode	Enum	Yes	Lunch / Dinner	Waiter
Session Start Time	Button	yes	—	System
Pause Session	Button	Yes	Temporarily stops listening	Waiter
End Session	Button	yes	Ends the session permanently	Waiter
Session Status	Enum	Auto	Started / Paused / Ended	System

System Behaviour

- Prevents multiple active sessions for the same table.
- Tracks session lifecycle (start, pause, resume, end).
- Associate the session with the waiter, outlet, and table.



Story 1.3 Live Session Experience

User Stories

- As a Waiter, I want real-time suggestions so I can respond confidently to guests & need guidance on tone and empathy without interrupting the conversation.
- As a Manager, I want menu priorities (specials, high-margin items) to influence AI recommendations.
- As a System, I want to detect allergies and flag unsafe recommendations.

During an active session, the system continuously listens to the conversation and provides real-time AI assistance.

A. Live AI Suggestions & Recommendations:

Data Captured by System (No Manual Input)

Data Element	Type
Audio Stream	Real-time
Speech-to-Text Transcript	Generated
Tone Indicators	AI-derived
Confidence Score	AI-derived
Empathy Score	AI-derived
Menu Query Context	AI-derived
Upsell Attempts	AI-detected

B. Tone & Behaviour Guidance (System Generated – Real-time)

Indicator	Display Type	Behaviour
Politeness Level	Colour bar/icon	Green / Amber / Red
Empathy Level	Score/icon	Real-time change
Confidence Level	Meter	Based on the voice tone
Difficult Situation Alert	Alert banner	Appears when guest dissatisfaction is detected

Example:

- “Try a softer tone”
- “Acknowledge guest concern”
- “Pause and reassure the guest”



C. Menu Recommendations Panel

Field	Type	Behaviour
Recommended Item	Card list	AI suggested
Reason for Recommendation	Text	Preference/allergy / ingredient
Taste Profile	Text	Spicy, mild, sweet
Allergy Warning	Alert tag	Highlighted if risk
Portion Size	Text	Light vs filling
Ingredient Insight	Tooltip	Ingredient-based info
Priority Tag	Badge	Special / High-margin

Logic Used by the System

- Guest preferences (from conversation)
- Allergy keywords
- Ingredient rules
- Manager-defined priorities

System Behaviour

- Continuously listens during an active session (opt-in visibility).
- Converts speech to text in real time.
- Analyzes:
 - Tone (polite, neutral, rude)
 - Confidence level
 - Conversation flow
 - Compliance with SOPs
- Generates **non-intrusive “ghost coach” prompts**:
 - Tone softening alerts
 - Empathy nudges
 - Upsell suggestions
 - Allergy warnings
- Uses **manager-defined menu priorities** to bias recommendations.
- Logs upsell attempt vs outcome.

D. Guest Feedback Capture

User Stories: As a Waiter, I want to capture quick feedback so I can understand guest satisfaction.



Field Name	Data Type	Mandatory	Validation
Star Rating	Integer	Yes	1–5
Session ID	UUID	Auto	—
Waiter ID	UUID	Auto	—

System Behaviour

- Prompts for rating only before session end.
- Stores feedback against the session.
- Uses rating in performance scoring and trend analysis.
- Simple star-based rating at the end of the session
- Captures the immediate guest satisfaction signal
- Linked to the specific session and waiter

End the session once the guest leave table

Story 1.4 Waiter Main Dashboard (Post-Session):

A. Performance metrics:

User Stories: As a Waiter, I want to see my performance so I know where to improve & I want clear indicators of missed opportunities.

Upsell & Performance Metrics – Definitions & Formulas

Metric Name	Definition	Formula / Calculation Logic	Notes / AI Logic Reference
Upsell Attempts	Total number of times an upsell recommendation was presented to a customer during an order interaction.	Upsell Attempts = Count of upsell prompts shown	Includes AI or staff-driven suggestions, regardless of customer acceptance.
Successful Upsells	Number of upsell attempts that resulted in the customer adding the recommended item(s) to the order.	Successful Upsells = Count of upsell items added to orders	Only counted when upsell item is successfully added and confirmed.
Upsell Success Rate	Percentage of upsell attempts that converted into successful upsells.	$(\text{Successful Upsells} / \text{Upsell Attempts}) \times 100$	Optional KPI for performance comparison.



Date: 12/01/2026

Missed Opportunities	Number of eligible upsell scenarios where no upsell was attempted.	Missed Opportunities = Eligible Upsell Scenarios – Upsell Attempts	Eligible scenarios identified by AI based on order context and menu rules.
Confidence	AI-evaluated score reflecting how confidently staff delivers upsell recommendations.	Confidence Score = Average AI confidence score per interaction	Based on speech clarity, hesitation, response timing, and assertiveness.
Menu Knowledge	Measure of staff accuracy in describing menu items, pricing, and combinations.	(Correct Menu Responses / Total Menu-related Interactions) × 100	Evaluated via AI validation against menu database.
Tone	AI-evaluated measure of friendliness, professionalism, and appropriateness during customer interaction.	Tone Score = Average tone quality score per interaction	Based on sentiment analysis, politeness, and non-pushy upsell delivery.

Metric
Total Upsell Attempts
Successful Upsells
Missed Upsell Opportunities
Confidence Score
Menu Knowledge Score
Tone Management Score
Empathy Score

System Behaviour

- Aggregates session-level data into daily/weekly metrics.
- Highlights weak areas visually.

B. Conversation History:

Field
Session Date & Time
Table Number
Guest Name



Date: 12/01/2026

Service Mode
Star Rating

On click (Session Details):

Performance Metrics Table

Metric Name	Score / Value	Description
Food Safety Awareness Score	% / Score	Allergy & safety compliance
Identified Upsell Opportunities	Count	Total opportunities detected
Empathy Score	Score (1–5)	Emotional handling
Tone Consistency	%	Stable polite tone
Menu Knowledge Accuracy	%	Correct menu info shared

System Behaviour

- Stores session summaries (not full audio unless configured).
- Allows read-only transcript view (optional).
- Masks sensitive guest data.

C. Leaderboard View (Peer Comparison)

User Stories: As a Waiter, I want to see where I stand among my peers & I want recognition when I perform well.

Fields Displayed (Waiter View)

Field	Description
Rank	Position
Waiter Name / Alias	Privacy controlled
Items Sold	Count
Progress %	Visual
Badge Earned	Icon

System Behaviour



- Sorts the leaderboard based on the selected metric.
- Handles tie-breakers using timestamp or secondary metrics.
- Refreshes ranking dynamically.

D. Learning & Coaching:

User Story: As a Waiter, I want short lessons so I can improve without long training sessions.

Field Name	Type	Description
Lesson ID	UUID	System generated
Lesson Title	Text	Visible to waiter
Lesson Category	Enum	Tone / Empathy / Menu / Upsell
YouTube Link	URL	Training video
Lesson Description	Text	What waiter will learn
Related Metric	Enum	Empathy / Tone / Menu
Assigned Outlet / Role	Multi-select	Target audience

System Logic (Auto Mapping)

Metric Condition	System Action
Low Empathy Score	Assign an empathy lesson
Low Tone Consistency	Assign a tone improvement lesson
Low Menu Accuracy	Assign the menu knowledge video
Missed Upsells	Assign the upselling techniques video

Waiter Side: My Learning Dashboard

Field	Type
Lesson Title	Text
Category	Tag
YouTube Link	Clickable
Completion Status	%



Assigned Reason	Metric-based
Completion Date	Date

System Behaviour (Learning)

- Tracks video link access
- Marks lesson completed (manual or % watched – configurable)
- Improves AI coaching suggestions after lesson completion
- Reflects improvement in future performance metrics

Waiter (Frontline User) Flow

User Journey

1. User logs into the mobile application using assigned credentials
2. User starts a live session when approaching a customer or guest
3. Application listens to the live conversation
4. Real-time transcript is generated (optional visibility)
5. AI provides live suggestions based on:
 - Menu questions
 - Allergies
 - Recommendations
 - Tone and empathy cues
6. User adapts responses using AI guidance
7. Session ends manually
8. Post-session summary is generated
9. User reviews performance, feedback, and badges
10. User is guided to micro-lessons if improvement is needed

Key Features

- Login / Logout
- Start & stop live conversation session
- Live transcript (read-only)
- Real-time AI suggestions (“ghost coach”)
- Tone and empathy alerts
- SOP-compliant response guidance
- Post-session performance summary
- KPI scores (tone, empathy, menu knowledge, upsells)
- Missed upsell indicators
- Badge earning & leaderboard visibility
- Micro-lesson recommendations



2. Restaurant Manager (Admin / Manager Role)

2.1 Signup & Login

User story: - as a manager, Super admin will add me in system, and I'll log in with credentials to the web-based admin dashboard

Login

Field	Type
Email ID/ Phone number	Email/ Number
Password	Password

System Behaviours

- System will validate email and phone uniqueness, & Applies role-based access (Manager/Admin).
- Logs login activity. Supports secure session handling.

2.2 Staff Management

User Stories: I want to create waiter accounts so they can use the mobile app, and I can track staff performance and training status.

Inputs / Fields (Create Staff)

Field	Type	Mandatory
Staff Name	Text	Yes
Email ID	Email	Yes
WhatsApp Number	Number	Yes
Role	Enum	Yes (Waiter)
Status	Enum	Active / Inactive

System Behaviours

- Sends invitation via Email / SMS / WhatsApp with app link and credentials.
- Links staff to the restaurant and outlet.
- Prevents deactivated staff from logging in.
- Tracks staff training completion and performance.



2.3 Restaurant Orientation & Data Setup (detailed down in 3. orientation epic)

Manager feeds and maintains all operational data required for AI training and guidance.

2.4 Manager Dashboard (Business & Performance View)

Date Range - Default View: Last 30 Days

A. ROI & Business Impact Summary (Read-only Metrics)

User Stories, I want to compare key performance metrics before and after platform adoption, so I can clearly measure operational impact.

CoreVista ROI – This Month

Area	Metric	Description
ROI	Additional Revenue	Revenue generated via AI
Comparison	Before vs After	Upsell rate, avg check size

Before vs After CoreVista Metrics

- Upsell Success Rate
- Average Order Size
- Customer Stratification
- Service Errors
- Staff Confidence Score

System Behaviours

- Calculates ROI automatically.
- Compares historical vs current data.
- Updates metrics periodically.

B. Revenue & Sales Analytics

User story - I want to see revenue broken down by food and beverage categories, so I can identify which categories benefit most from AI-driven upselling.

Area	Feature	Description
Category Analysis	Revenue by Category	Wine, pasta, desserts
Item Analysis	Item-wise Sales	Visual bar charts



Errors	Service Errors	Missed SOPs, mistakes
--------	----------------	-----------------------

System Behaviours

- Aggregates item-level sales into categories
- Displays bar or stacked charts
- Allows category-wise comparison across time periods

2.5 Staff Performance List & Drill-Down View

User Story - I want to drill down into individual staff performance, so I can provide personalised feedback and coaching.

Metric Name	Definition	Formula / Calculation Logic	Notes / AI Logic Reference
Upsell Attempts	Total number of times an upsell recommendation was presented to a customer during an order interaction.	$\text{Upsell Attempts} = \text{Count of upsell prompts shown}$	Includes AI or staff-driven suggestions, regardless of customer acceptance.
Successful Upsells	Number of upsell attempts that resulted in the customer adding the recommended item(s) to the order.	$\text{Successful Upsells} = \text{Count of upsell items added to orders}$	Only counted when upsell item is successfully added and confirmed.
Upsell Success Rate	Percentage of upsell attempts that converted into successful upsells.	$(\text{Successful Upsells} / \text{Upsell Attempts}) \times 100$	Optional KPI for performance comparison.
Missed Opportunities	Number of eligible upsell scenarios where no upsell was attempted.	$\text{Missed Opportunities} = \text{Eligible Upsell Scenarios} - \text{Upsell Attempts}$	Eligible scenarios identified by AI based on order context and menu rules.
Confidence	AI-evaluated score reflecting how confidently staff delivers upsell recommendations.	$\text{Confidence Score} = \text{Average AI confidence score per interaction}$	Based on speech clarity, hesitation, response timing, and assertiveness.
Menu Knowledge	Measure of staff accuracy in describing menu items, pricing, and combinations.	$(\text{Correct Menu Responses} / \text{Total Menu-related Interactions}) \times 100$	Evaluated via AI validation against menu database.
Tone	AI-evaluated measure of friendliness, professionalism, and appropriateness during customer interaction.	$\text{Tone Score} = \text{Average tone quality score per interaction}$	Based on sentiment analysis, politeness, and non-pushy upsell delivery.



Staff Metrics

Metric
Upsell Attempts
Successful Upsells
Missed Opportunities
Confidence
Menu Knowledge
Tone

System Behaviour: Pulls aggregated metrics per staff & shows trends over time. Highlights weak and strong areas

Conversation-Level View

User story - I want to see conversation-level details, so I can audit specific interactions if needed.

Field
Table Number
Guest Name
Service Mode
Session Date

System Behaviour- Lists all sessions chronologically & allows filtering and searching. Opens the session detail view on selection

Session Detail Metrics (Table)

User story - I want detailed session metrics, so I can understand exactly what happened during an interaction.

Metric Name	Score / Value Type	Definition	Formula / Calculation Logic	Notes / AI Logic Reference
-------------	--------------------	------------	-----------------------------	----------------------------



Food Safety Awareness	% / Score	Measures staff compliance with food safety and allergy-related communication during interactions.	$(\text{Correct Safety-Compliant Interactions} / \text{Total Safety-related Interactions}) \times 100$	Includes allergy disclosure, cross-contamination warnings, and safety confirmations detected by AI.
Upsell Opportunities	Count	Total number of valid upsell opportunities identified during customer interactions.	$\text{Upsell Opportunities} = \text{Count of AI-detected eligible upsell scenarios}$	Identified based on order context, menu rules, and timing.
Empathy Score	Score (1–5 or %)	Measures how effectively staff recognizes and responds to customer emotions or special needs.	$\text{Empathy Score} = \text{Average AI empathy rating per interaction}$	Based on acknowledgment, reassurance, patience, and emotional alignment.
Tone Consistency	%	Measures how consistently a polite and professional tone is maintained throughout interactions.	$(\text{Interactions with Consistent Tone} / \text{Total Interactions}) \times 100$	AI tracks tone stability and detects abrupt or negative shifts.
Menu Knowledge Accuracy	%	Accuracy of menu-related information shared with customers.	$(\text{Correct Menu Information Instances} / \text{Total Menu Information Instances}) \times 100$	Validated against menu database (items, ingredients, pricing, availability).
Guest Rating	Score (1–5) / %	Average customer feedback rating provided post-interaction or post-visit.	$\text{Guest Rating} = \text{Average of customer ratings received}$	May be sourced from app, SMS, QR, or POS feedback systems.

Metric
Food Safety Awareness
Upsell Opportunities



Date: 12/01/2026

Empathy Score
Tone Consistency
Menu Knowledge Accuracy
Guest Rating

System Behaviour

- Evaluates session transcript post-call
- Scores each metric using AI + rules
- Stores results for reporting and coaching

2.6 Knowledge & Coaching Management

User story - I want to create and manage coaching lessons, so I can continuously improve staff skills.

Inputs / Fields

Field	Type
Lesson Title	Text
Lesson Type	Enum
YouTube Link	URL
Mapped Skill	Enum
Assigned Staff	Multi-select
Completion %	Auto

System Behaviour

- Stores lessons in the knowledge base
- Maps lessons to skills and KPIs
- Tracks staff completion automatically
- Feeds lessons into AI recommendation engine

Restaurant Manager (Admin / Manager Role) Flow

User Journey

1. Manager logs into the web-based admin dashboard



2. Sets up restaurant profile and operational data
3. Uploads menu, ingredients, SOPs, and policies
4. Defines KPIs, grading scales, and service benchmarks
5. Creates and manages waiter/receptionist accounts
6. Monitors live and historical performance data
7. Reviews post-session summaries and analytics
8. Identifies training gaps and coaching needs
9. Configures sales goals and gamification rules
10. Reviews ROI, revenue impact, and engagement metrics

Key Features

- Login / Logout
- User creation and role management
- Menu upload (PDF → structured text)
- Ingredient & allergy management
- SOP and policy configuration
- KPI creation and weighting
- Grading scale definition (% / score / units)
- Dashboard of all users and sessions
- Performance analytics (time, upsells, missed opportunities)
- Staff confidence and coaching metrics
- Micro-lesson content management

3.) Orientation (Data needs to be feed by the Manager)

Module 3.1: Restaurant Standard Policies (MANDATORY)

User Stories

- As a Manager, I want to define standard policies so staff give uniform answers.
- As a System, I want one single source of truth for policy-related queries.
- As a Waiter, I want confidence while answering policy questions.

Field	Exact Details to Cover	System Behavior
Operating Timings	Opening time, closing time, last order time	Single source of truth shown to all staff
Waiting Policy	Walk-in waiting rules, queue handling	Prevents inconsistent answers
Reservation Policy	Booking rules, no-show, cancellation	Referenced during guest queries
Table Holding	How long a table can be held	Enforced during seating
Dining Rules	Dine-in vs takeaway, outside food	Staff cannot override
Guest Accommodation	Child seating, elderly, wheelchair	Prompt shown during seating



Payments	Cash, card, UPI, split bills	Displayed during billing
----------	------------------------------	--------------------------

System Behaviour

- Uses these rules during live guest conversations.
- Prevents contradictory responses across staff.
- Flags policy violations during sessions.
- Displays prompts when policy-related questions arise.

Module 3.2: Menu Knowledge (MANDATORY)

User Stories

- **As a Manager**, I want accurate menu data to avoid misinformation.
- **As a System**, I want allergy-safe recommendations.
- **As a Waiter**, I want confidence while answering menu questions.

Field	Details	Behavior
Dish Type	Veg / Non-veg	Trust building
Taste Profile	Spicy, mild, sweet	Confident answers
Ingredients	all	Allergy awareness
Portion Size	Light vs filling	Expectation setting
Popular Items	Best sellers, specials	Recommendation readiness

System Behaviour

- Uses ingredient data for **allergy detection**.
- Powers real-time menu recommendations.
- Flags incorrect menu knowledge during sessions.
- Scores waiter accuracy for training insights.

Module 3.3: Service SOP – Flow of Service (MANDATORY)

User Stories

- **As a Manager**, I want a defined service flow to maintain brand standards.
- **As a System**, I want to detect SOP deviations.
- **As a Waiter**, I want guidance to avoid mistakes.



Step	Description	Outcome
Greeting	Warm welcome	First impression
Seating	Preference-based	Comfort
Menu Handover	Clear & calm	Clarity
Order Taking	Listen & repeat	Error reduction
Order Confirmation	Verify preferences	Trust
Serving	Timely & accurate	Satisfaction
Table Check	Once post-serving	Care without hovering
Clearing	Polite	Closure
Billing	Transparent	Smooth end
Farewell	Thank & invite back	Brand recall

System Behaviour

- Tracks SOP compliance during sessions.
- Flags missed or skipped steps.
- Uses SOP adherence in performance scoring.
- Suggests corrections in real time.

Module 3.4: Communication, Tone (MANDATORY)

User Stories

- **As a Manager**, I want tone standards to reduce complaints.
- **As a System**, I want to detect tone deviations automatically.

Aspect	Trained Behavior	Purpose
Tone	Calm, respectful	De-escalation
Language	Polite phrasing	Professionalism
Listening	No interruption	Guest feels heard

System Behaviour

- Analyzes voice tone and speech patterns.



- Triggers alerts for rude or dismissive behavior.
- Scores tone consistency per session.

Module 3.5: Best Practices & Excellence (ADVANCED)

User Stories

- **As a Manager**, I want to define excellence standards beyond basics.
- **As a System**, I want to coach high-performing staff further.

Area	Focus
Anticipation	Water, napkins, cutlery
Peak Hours	Speed + clarity
VIPs	Personal touch
Recovery	Calm mistake handling

System Behaviour

- Applies advanced coaching to experienced staff.
- Rewards proactive service with higher scores.
- Identifies staff ready for leadership roles.

Module 3.6 Sales Goals & Campaigns

Field	Type
Goal Name	Text
Goal Type	Daily / Weekly
Target Items	Multi-select
Target Value	Number
Validity Period	Date range



4. System Output (AI & Platform Logic)

4.1: Conversation Processing Engine

Input	Source
Live Audio Stream	Waiter mobile app
Session Metadata	Table, waiter, time

Features- core function

- Audio ingestion
- Speaker diarization
- Speech-to-text (Deepgram)
- Real-time processing pipeline

System Behaviour

- Captures live audio during active sessions only
- Identifies **waiter vs guest** using speaker diarization
- Converts speech to text in near real time (Deepgram / Whisper)
- Streams structured text downstream for analysis
- Stops processing on session pause/end

4.2: Intent, Emotion & Tone Engine

Input
Transcribed conversation
Speaker labels
Orientation rules (tone, SOPs)

Features

- Intent detection (allergy, pricing, upsell)
- Sentiment classification
- Tone analysis
- Conversation temperature detection
- Trend analysis over session

System Behaviour



- Detects **intent** such as:
 - Allergy inquiry
 - Pricing question
 - Recommendation request
 - Complaint
 - Upsell opportunity
- Classifies **customer sentiment**:
 - Positive
 - Neutral
 - Frustrated / Negative
- Evaluates **waiter tone**:
 - Polite
 - Empathetic
 - Confident
 - Rushed / defensive
- Tracks **conversation temperature** over time:
 - Calm
 - Neutral
 - Sensitive
 - Escalating
- Sends real-time alerts to the waiter app when risk is detected

4.3: RAG Knowledge Engine

Input
Manager-uploaded data
Detected intent
Conversation context

Features- Core Functions

- Vector embedding generation
- Real-time data retrieval
- Zero retraining architecture
- Dynamic updates from manager uploads

System Behaviour

- Converts all manager-fed data (menu, SOPs, policies) into embeddings
- Stores embeddings in a vector database
- Fetches **only relevant data** based on live intent
- Ensures:
 - No model retraining required
 - Real-time updates when manager changes data



- Acts as the **single knowledge source** for AI responses

4.4: SOP & KPI Evaluation Engine

Input
Session transcript
SOP definitions
KPI configurations

Features -core function

- Rule-based checks
- LLM-based reasoning
- KPI scoring per session
- Highlight extraction
- Feedback generation

System Behaviour

- Evaluates conversations against:
 - Greeting SOPs
 - Menu accuracy rules
 - Allergy handling rules
 - Tone & professionalism standards
- Calculates **KPI scores per session**
- Identifies:
 - Missed SOP steps
 - Incorrect responses
 - Missed upsell opportunities
- Generates structured feedback for waiter and manager dashboards

4.5: Recommendation & Learning Engine

Input
KPI scores
Coaching triggers
Learning content (YouTube links)
Gamification rules

Features -core function



- Micro-lesson recommendation
- Coaching trigger detection
- Continuous improvement loop
- Gamification logic

System Behaviour

- Maps weak KPIs to relevant micro-lessons
- Assigns lessons automatically to staff
- Adjusts future coaching prompts based on lesson completion
- Improves AI guidance over time using performance trends

4. 6 Order Confirmation Prompts

- System displays a *Confirm Order* view

Post-Session System Outputs

- Session summary generation
- KPI-wise scoring
- Missed opportunity analysis
- Coaching and learning recommendations
- Data aggregation for manager dashboards and ROI analytics

System Behaviour

- Detects when an order is placed
- Displays **Confirm Order View** on waiter app
- Prompts waiter to:
 - Repeat order
 - Confirm preferences
 - Verify allergies
- Flags skipped confirmation as SOP deviation
- Reduces order errors and disputes



Phase- 2:

2.5 AI Coaching & Learning Insights

User story — I want a consolidated view of AI coaching effectiveness, staff learning progress, and AI-recommended actions, so that I can understand whether coaching is improving performance, track skill adoption over time, and take timely, data-driven actions to improve service quality and business outcomes.

A. Coaching Effectiveness (System Metrics)

Metric
Most Triggered Scenarios
Response within 30 seconds
Successful Outcomes
Avg Implementation Time

B. Learning Progress & Adoption

Metric
Staff requiring less coaching over time
Avg Days to Proficiency
New Hire Ramp-up Time

C. AI Recommended Actions

Recommendation
Focus on low performers
Recognition suggestions for top performers
Identify peak hours
Highlight priority coaching areas



Date: 12/01/2026

2.6 Overall Impact Summary

User story – I want a consolidated impact summary, so I can quickly understand platform-wide outcomes.

Metrics

Field
Total Coaching Sessions
Estimated Turnover Savings
Staff Satisfaction Indicators

System behaviour

Aggregates data across all modules & Calculates savings based on reduced errors and improved retention.
Displays high-level KPIs in a summary card view